Yongxi Feng (冯泳熙)

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Personal Website LinkedIn GitHub Languages: Mandarin Chinese (native) | Cantonese (native) | English (fluent)

Note: Underlined entries are hyperlinked to supporting documentation, code repositories, or official records.

Education

The Chinese University of Hong Kong, Shenzhen (CUHK-SZ), *Shenzhen, China*

B.S. in Statistics *August 2023 – June 2027*

CGPA: 3.879/4.0 Major GPA: 3.926/4.0 Rank: 2/67 (major), 18/361 (School of Data Science)

- **Dean's List Award**: AY2023–24, AY2024–25, AY2025–26 (three consecutive years)
- **Academic Performance Scholarship**: Class C (AY2024–25); Class A (AY2025–26)
- Core Courses: Honors Probability & Statistics, Linear Models, Machine Learning, Reinforcement Learning, Time Series Analysis, Stochastic Processes, Optimization, Mathematical Analysis

University of British Columbia, Faculty of Science, *Vancouver, BC, Canada*

Exchange Student *January 2026 – April 2026*

- Coursework: Time Series Analysis (relevant to ongoing wearable data research)

Research Experience

Wearable Sensor Anomaly Detection —LLM-Based Framework, *CUHK-Shenzhen*

Undergraduate Researcher *(Advisor: Prof. Kaixin Zhou)* *Summer 2025*

- Investigated the clinical validity gap in wearable health data: existing anomaly detection approaches rely on population-level statistical thresholds that conflate device artifacts with subject-specific physiological deviations
- Processed and tokenized time-series physiological signals from **NHANES** (2,000+ participants) into structured sequences suitable as LLM inputs; designed prompt engineering and fine-tuning strategies to adapt LLMs for anomaly detection over irregular temporal data
- Implemented a teacher-student framework with knowledge distillation, using FRIM-based statistical soft labels to jointly learn population-level and individualized anomaly patterns
- Benchmarked proposed approach against statistical and deep learning baselines, demonstrating robustness in detecting irregular activity patterns across heterogeneous participants

Decision Trajectory Modeling: Temporal Point Processes + Diffusion Models, *CUHK-Shenzhen*

Undergraduate Researcher *(Advisor: Prof. Shuang Li)* *2025 – Present*

- Investigated hybrid probabilistic-generative framework for modeling when and how human decisions evolve: standard TPPs capture event timing; diffusion models capture semantic evolution; the joint problem remains open
- Designed dual-time architecture separating real time (event intensity via Hawkes processes) from diffusion time (latent noise evolution), using an LLM as the score network for hazard-guided, context-aware trajectory generation
- Contributed to paper conceptualization and preliminary model design in collaboration with graduate researchers; paper title: *"HAZARDS GUIDE NOISE: LLM-Conditioned Diffusion with Temporal Point Processes for Human Decision Trajectories"*

"Photo Stories": Visual-Guided Interview & Narrative Generation System, *Tencent Collaboration*

Research Engineer *(Advisor: Prof. Zhongxiang Dai)* **[GitHub]** *2025 – Present*

- Built multimodal system enabling visually grounded oral history collection: users upload photographs; the system analyzes visual content (objects, architectural details, etc.) and generates contextually targeted interview questions
- Designed and implemented the **question generation pipeline**: extracting semantically meaningful visual features and mapping them to coherent interview prompts; developed cross-image context management to enable longitudinal narrative construction
- Developed simulation interview corpus for system evaluation; conducted **benchmark evaluation** across multiple LLMs (Gemini, Hunyuan, Tencent Yuanbao) on question relevance, narrative coherence, and story richness
- Constructed RAG-enhanced demo for live interaction; pipeline extends to automated narrative synthesis combining visual descriptions with collected oral content

Seasonal Disease Patterns During Spring Festival: Clinical Distribution Study, ***Minerva College, CUHK-SZ***

Lead Data Analyst *Spring 2025*

- Collaborated with clinical medicine students to analyze case data from five tertiary hospitals, examining seasonal and departmental patterns of high-incidence diseases across regions
- Processed and structured real-world hospital data; conducted semi-structured interviews with clinicians to integrate domain knowledge into statistical interpretation and public health recommendations
- Co-developed prevention and resource allocation recommendations; presented findings via academic poster and public science communication event
- **Outstanding Capstone Project Award**, Minerva College, CUHK-Shenzhen **[Verification]**

Independent Study: Deep Active Inference & Cognitive Modeling, ***CUHK-Shenzhen***

Participant *(Organized by Dean Jianhua Huang, School of AI; co-studied with PhD mentors)* *2024 – 2025*

- Selected for faculty-initiated reading and research group; studied foundational literature at the intersection of active inference, the Free Energy Principle, and deep learning architectures for cognitive modeling
- Delivered literature presentations and contributed to paper draft: *"Deep Active Inference: Bridging the Free Energy Principle with Cognitive Modeling in Deep Learning Architectures"*

Teaching & Academic Service

Undergraduate Teaching Assistant (USTF), MAT3007 Optimization, ***CUHK-Shenzhen***

(Instructor: Prof. Xiao Li) **[Verification]** *Fall 2025*

- Conducted three tutorial sessions decomposing convex optimization concepts for mixed-background cohorts; held weekly office hours with individualized problem-solving support; evaluated assignments and provided written feedback

Peer Academic Advisor, School of Data Science, ***CUHK-Shenzhen***

(Coordinator: Prof. Dan Li) **[Verification]** *AY2025–2026*

- Selected as outstanding peer advisor; conducted one-on-one advising for 20+ students on academic planning, research entry, and graduate application strategy; organized information sessions on major pathways

Panelist, Undergraduate Academic Exchange Session, ***CUHK-Shenzhen SDS***

(Referee: Prof. Shuang Li) **[Event]** **[WeChat]** *Fall 2025*

- Selected as student speaker to share experiences in research entry and study abroad

Review Session Instructor, MAT3007 Optimization, ***Minerva College, CUHK-SZ***

- Initiated and delivered exam preparation lectures in coordination with the residential college *Spring 2024*

Volunteer English Teacher, Rural Primary School, ***Guangxi, China***

- Designed and delivered English curriculum for students with heterogeneous proficiency levels *Spring 2023*

Selected Course Projects

Time Series (STAT443): *Forecasting Daily Influenza Incidence in Kawasaki City* (2014–2025) — Compared AR(1), SARIMA, and ARIMAX models under one-step-ahead framework; showed strong endogenous temporal structure limits marginal gains from meteorological covariates.

Machine Learning (DDA3020): *Hybrid ML Trading Strategy for EMH Testing* (Kaggle Competition) — Decoupled prediction from execution: LightGBM for directional signals + PPO reinforcement learning agent for volatility-adaptive position sizing; demonstrated consistent excess returns over S&P 500 across market regimes.

Reinforcement Learning (DDA4230): *Credit Assignment in Closed Cooperative Systems* — Systematic survey of MARL credit assignment paradigms (VDN, QMIX, QTRAN, COMA, MADDPG, TarMAC); analyzed theoretical assumptions and practical trade-offs in cooperative settings.

Technical Skills

Programming & Tools

- Python (PyTorch, scikit-learn, pandas, NumPy)
- R (tidyverse, survival, ggplot2)
- SQL, MATLAB, Stata, LaTeX
- Git / GitHub

Methods

- Statistical learning, time-series analysis
- Multiple imputation, missing data methods
- Reinforcement learning, LLM fine-tuning
- Uncertainty quantification (introductory)

Research Interests: Machine learning reliability and clinical validity in healthcare data; wearable/sensor time-series modeling; uncertainty quantification; bridging algorithmic anomaly detection and clinical interpretation.